

AI-Powered Agricultural Advisory System for Smart Farming Using Machine Learning

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Abstract— Agriculture is very important for providing food and supporting the country's economy. However, farmers face many difficulties such as plant diseases, difficulty in selecting the right crops, and frequent changes in market prices. Most traditional farming methods depend on personal experience and manual checking, which takes a lot of time and can sometimes result in incorrect decisions.

This project presents an agricultural advisory system that helps farmers by using different computer-based models to support their daily farming activities. The system brings together crop recommendation, plant disease detection, and market price prediction on a single platform. For detecting plant diseases from images, Convolutional Neural Network (CNN) models are used. Random Forest algorithms help in suggesting suitable crops based on soil type and environmental conditions. Long Short-Term Memory (LSTM) models are used to predict future market price patterns.

The performance of these models is checked using measures such as accuracy and reliability to find the best-performing methods. The system also provides timely suggestions through an easy-to-use interface, making it simple for farmers to understand and use. The results show that this system can help increase crop production, reduce losses caused by diseases, and support better farming decisions based on available data.

Keywords— AI-Powered Agriculture, Crop Recommendation, Disease Detection, Price Prediction, Convolutional Neural Networks, Random Forest, LSTM, Smart Farming

I. INTRODUCTION

Agriculture is one of the most important section in the world as it gives food and supports the country. Crops like rice, wheat, maize, and potatoes are majorly grown to meet the food needs of the increasing population. These crops can be cultivated in different types of climates, ranging from cool temperate areas to hot tropical areas, which makes them important for supporting people's livelihoods and contributing to the growth of country. They are also nutritious and provide essential nutrients such as carbohydrates, vitamins, and minerals that are necessary for maintaining good health. In addition, these crops grow in a short time and produce good yields, which makes them helpful in managing food shortages and the increasing population. Because of this, increasing agricultural productivity has become very important to ensure food security in the future [1]. The growth and production of crops largely depend on detecting diseases at an early stage and identifying them correctly. Traditional methods of detecting plant diseases mainly depend on manual checking by farmers. However, this method is often slow, may give different results, and can lead to mistakes. These methods do not provide the level of accuracy needed in modern

farming, making it difficult to identify diseases on time and take proper preventive actions [2].

Inefficient farming methods and lack of proper management are some of the main reasons behind many agricultural problems. Increasing crop yield is important, but maintaining the quality of crops is also necessary for sustainable farming. However, traditional farming techniques often lead to lower efficiency and productivity because they mostly depend on manual decisions. Agricultural production is also affected by factors such as weather conditions, soil quality, and proper disease management. To deal with these challenges, modern technologies like Artificial Intelligence (AI) and the Internet of Things (IoT) are increasingly being used in farming. These technologies help farmers monitor farm conditions automatically, identify diseases at an early stage, and use resources more efficiently. At the same time, it is important to follow eco-friendly farming processes that decreases damage to the environment while increasing overall productivity [3].

Crop diseases and improper crop selection are major challenges faced by farmers and often lead to heavy losses in both yield and quality. Many plant diseases are caused by fungal, bacterial, and viral infections, which can affect crops at different stages of growth. Because of this, early detection of diseases is very important for proper control [4]. In addition, choosing the wrong crop based on unsuitable soil or environmental conditions can further reduce productivity. Changes in market prices also create uncertainty, making it hard for farmers to decide the best time to sell their crops. To deal with these challenges, there is a need to develop systems that can provide reliable suggestions for disease detection, crop selection, and market trends. This study aims on developing a system that uses image-based methods to identify plant diseases and combines them with other advisory features to help farmers make better decisions [5]

Recent developments in machine learning and deep learning have made it easier to analyze agricultural data more accurately. Methods such as Convolutional Neural Networks (CNNs) are commonly used to detect plant diseases from leaf images by identifying signs like color changes, spots, and texture differences. Machine learning models such as Random Forest are useful for suggesting suitable crops based on soil nutrients and weather conditions. Similarly, real-time models like Long Short-Term Memory (LSTM) are used to predict future market prices by evaluating past data. These technologies help in creating smart models that support farmers in making timely and better decisions [6]. However, many existing systems focus on only one task, such as disease detection, and do not provide complete support to farmers. Therefore, there is a need to create an integrated models that combines multiple features on a single platform to make farming more efficient and easier to manage [7].

Deep learning has become an important tool in modern agriculture because it helps in the early detection of plant diseases and reduces crop losses while improving overall productivity. In this study, different deep learning models such as custom CNN, VGG16, ResNet50, and InceptionV3 are studied to check their performance in identifying plant diseases using image-based data. These models are trained using leaf image datasets, and their performance is checked using measures such as accuracy and overall classification results [8]. Understanding the main causes of plant diseases is very crucial for proper crop management. While deep learning models make disease detection easier through image analysis, other factors such as soil quality, weather conditions, and pest control methods also affect crop health. However, many existing processes do not provide clear recommendations about the correct use of pesticides. Therefore, there is a need for a modern system that considers different factors like crop type, soil nutrients, weather conditions, and pest levels. This helps farmers make better decisions and manage their crops more effectively [9]–[10].

The project focuses on creating an agricultural advisory model that is easy for farmers to use, affordable, and able to provide accurate suggestions. By using models like custom CNN, Random Forest, and LSTM, the system can detect plant diseases at an early phase, suggest suitable crops based on soil conditions, and predict market price trends. This helps farmers make timely and informed decisions. In the future, the system can be integrated into a mobile or cloud-based platform, making advisory services faster and easier to use. The platform can also provide particular recommendations for fertilizers and pesticides, ensuring proper treatment while reducing the unnecessary use of chemicals. This approach helps farmers take early step and make better decisions, supporting sustainable farming, improving crop quality, and reducing overall loss.

LITERATURE REVIEW

In present years, many researchers have studied the use of machine learning and deep learning techniques to enhance decision-making in agriculture. Singh et al. [11] proposed a structured process for accurate plant disease detection using the well-known PlantVillage dataset. Their method included image segmentation using K-means clustering to identify important areas of leaf images, followed by feature extraction using the Gray-Level Co-occurrence Matrix (GLCM) to capture texture details. After that, a multiclass Support Vector Machine (SVM) classifier was used to classify different diseases. This process achieved a high accuracy of 95.99%, showing that combining image processing and machine learning techniques can be very effective.

Similarly, Demilie et al. [12] conducted a large-scale study that focused on the rapid development of deep learning methods and highlighted their increasing importance in agricultural applications. Arunodai et al. [13]

pointed out that deep learning models have made it much easier to detect plant diseases and pests when compared to traditional manual methods. In their study, the ResNet-50 model was used and showed good results in identifying pests such as locusts and larvae through image analysis. The process included several stages, including cleaning the data, preparing the images, segmenting important parts, and training the model to improve detection accuracy. Apart from disease detection, modern agricultural systems are also designed to provide guidance on pesticide usage. This helps protect crops while also reducing the excessive use of harmful chemicals.

In addition to this, researchers have created crop suggestion systems using machine learning methods such as SVM, Decision Tree, and Logistic Regression. Among these techniques, SVM produced better results than the others in many cases. These systems help farmers select crops that match soil and environmental conditions, which leads to better planning and higher productivity. Kumar et al. [14] also highlighted that improving training methods plays an important role in increasing model performance, especially in detecting foliar diseases. Their study showed that regular enhancement in deep learning models can increase detection accuracy and support the growth of healthier crops with improved yields.

Agriculture plays an important role in maintaining economic stability in many areas. However, crop production is often affected by difficulties like plant diseases, pest attacks, and changing environmental conditions, which can decrease the yield and farmers' income. Traditional methods of disease detection, such as manual inspection and field observation, are usually slow and not very effective for large-scale farming. To overcome these problems, Bagban et al. [15] developed automated systems that combine image processing with machine learning models like SVM and CNN. These systems analyze leaf features such as color and texture to detect diseases quickly. As a result, farmers can identify problems at an early stage and take timely action to reduce crop losses. Islam et al. [16] also showed that enhancement in imaging technology and computing processes have made plant disease detection faster and more reliable. In their study, the PlantVillage dataset was used to train models capable of identifying different plant diseases with an accuracy of about 95%. This approach provides a practical and scalable solution that can be applied in real farming conditions. Similarly, Asif et al. [17] used CNN-based models to classify plant leaf images into healthy and diseased categories. By using transfer learning techniques, their model achieved an accuracy of up to 97%. They also used image segmentation methods to separate infected areas from healthy parts, which helped improve the overall classification results.

Furthermore, He et al. [18] introduced deep residual networks (ResNet), which improved image recognition performance by solving the vanishing gradient problem in deep neural networks. Their work also discussed feature extraction methods like VLAD and Fisher Vectors, which help represent image data more correctly. These developments have played an important role in creating more accurate and powerful models. Overall, combining these techniques in agricultural advisory systems not only supports disease detection but also helps with crop recommendation and prediction tasks. This allows farmers to make better decisions, increase productivity, and follow more sustainable farming practices.

I. PROPOSED METHODOLOGY

The proposed model makes use of artificial intelligence and deep learning techniques to create an integrated agricultural advisory system. It is designed to support farmers by offering solutions such as plant disease detection, crop suggestion and market price prediction. The overall method follows a step-by-step process that includes data collection, preprocessing, data augmentation, model training, and finally generating useful recommendations. Each of these stages plays an important role in improving the accuracy of the system and making it suitable for real-world use. By combining different AI models into a single framework, the system aims to provide reliable, affordable, and practical support for modern farming needs.

A. Data Collection and Preparation

The data used in this study was taken from publicly available agricultural datasets such as the Plant Village dataset and other open sources. It includes plant leaf images that are used for detecting diseases, soil-related details for suggesting suitable crops, and past market price data for making predictions. The image dataset contains multiple types of samples, that consists healthy leaves and leaves affected by various diseases.

Since the data was collected from different sources, there were differences in image quality, lighting, and background conditions. To manage these differences, a data cleaning stage was carried out. During this process, duplicate, unnecessary, and low-quality samples were removed. This helped make the dataset more reliable and useful for training machine learning models.

B. Image Preprocessing

To manage the dataset for use in deep learning models, a preprocessing stage was carried out to make the data more consistent. First, all images were resized to a standard size of $128 \times 128 \times 3$ pixels, so the models could manage them easily. However, in the case of the InceptionV3 model, the images were resized to $299 \times 299 \times 3$ pixels as this size is provided for that model to work properly.

Next, the pixel digits were valued to a range between 0 and 1, which helped the training process run more easily. Some noise reduction filters were also used to remove unwanted distortions from the images. This made the disease patterns easier to notice. After completing these steps, the dataset became more uniform and easier to work with, which helped the model give better classification results.

C. Data Augmentation for Improved Generalization

Data augmentation was used to make the deep learning models more accurate and able to manage different types of images. Instead of gathering new data, extra image variations were created from the existing dataset. This was done by applying methods like rotation, flipping, scaling, and shearing to the images. These changes helped increase the size and diversity of the dataset.

Brightness and contrast were also adjusted to represent images taken under different environmental conditions. Because of this, the model learned to recognize disease patterns even when the lighting, angle, or background was different. Overall, these steps helped the model perform better in real-world situations and made the system more dependable for agricultural use.

Selection for Deep Learning Models

To classify potato leaf diseases, four deep learning models were selected: Custom CNN, VGG16, ResNet50, and InceptionV3. These models were chosen because each one has different strengths in image identification tasks. The Custom CNN model was specially designed for this dataset, making it simple, lightweight, and efficient to use.

The VGG16 model is known for its strong ability to extract important features from images, while ResNet50 uses residual connections that help solve common problems in very deep networks. On the other side, InceptionV3 can capture image details at different scales, which helps in identifying difficult patterns. These models were selected so their performance could be compared in detecting diseases in potato plant leaves.

D. Crop Recommendation System

Along with disease detection, the system also includes a crop recommendation module that uses machine learning techniques. In this part, the Random Forest algorithm is applied to analyze soil properties such as nitrogen, phosphorus, potassium levels, and pH value. Environmental factors like temperature and humidity are also taken into account.

Using this details, the system suggest crops that are suitable for the given soil and weather conditions. This makes it easier for farmers to select the right crops for their land and helps decrease losses that can occur when crops are not suitable for the soil.

E. Market Price Prediction

To help farmers make better selling decisions, a price prediction module is included in the system. In this module, the Long Short-Term Memory (LSTM) model is used to study past market price data and understand how prices change over time. From these patterns, the system predicts possible future price trends.

By using this information, farmers can decide a better time to sell their crops rather than selling them without planning. This decrease the chances of losses caused by sudden price changes and helps farmers manage their sales in a more organized and profitable manner.

F. Model Training and Evaluation

All the selected models were trained using labeled datasets, and their performance was checked using measures such as accuracy, precision, recall, and F1-score. The dataset was divided into training and testing parts so that the models could be tested properly. Hyperparameter tuning was also done to improve the performance of the models.

After training, the outcome of all models were compared to find out which model performed the best for each work. This comparison helped in selecting the most suitable model, making the system more dependable for practical use.

The system processes the input using trained models and provides real-time recommendations. This integrated approach reduces the need for multiple tools and simplifies decision-making for farmers.

G. Pesticide Recommendation System

After detecting a plant disease, the next step is to provide proper treatment suggestions to farmers. Based on the detected disease, the system recommends suitable pesticides and fertilizers that can help control the problem. Instead of giving general solutions, the system looks at different factors such as crop type, level of disease severity, and environmental conditions before making recommendations.

For fungal infections, the system suggests commonly used fungicides, such as those containing mancozeb, copper-based compounds, or other preventive chemicals that help control the spread at an early stage. If the infection becomes severe, stronger treatments are suggested to provide deeper and longer protection. By combining disease detection with quick treatment suggestions, the system provides useful and timely support to farmers. This method helps farmers take action quickly, reduces the unnecessary use of chemicals, and supports better crop management practices.

II. DEEP LEARNING MODELS FOR DISEASE CLASSIFICATION

A. Custom CNN model

The custom CNN model was used as an effective method for identifying diseases such as late blight and early blight in potato leaves. This model consists of two convolutional layers with ReLU activation, followed by max pooling layers. These layers decrease the complexity of the data while focusing on the key features of the images. The extracted features are flattened and passed through a fully connected dense layer. In the final phase, a softmax output layer produced probability values for each disease category. The architecture shown in Fig. 1 is suitable for real-time applications, as it provides a good balance between accuracy and computational efficiency.

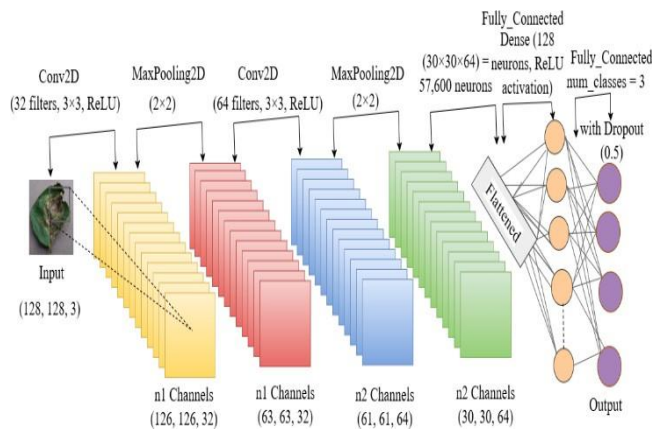


Fig. 1. Custom CNN Architecture

The process starts when an input image is uploaded to the system. The image first passes through a Conv2D layer that uses 32 filters of size 3×3 along with ReLU activation. At this stage, the model starts identifying basic patterns such as edges and textures. After that, a MaxPooling2D layer of size 2×2 reduces the image dimensions while keeping the most important information. essential features. The Next, the image moves to a second Conv2D layer with 64 filters of size 3×3 and ReLU activation, allowing the model to recognize

more detailed patterns. Another MaxPooling2D layer (2×2) is applied to further decrease the size of the feature maps, making the data easier to process. After finishing these steps, the model uses the extracted features to differentiate the leaf image and recognise the disease correctly.

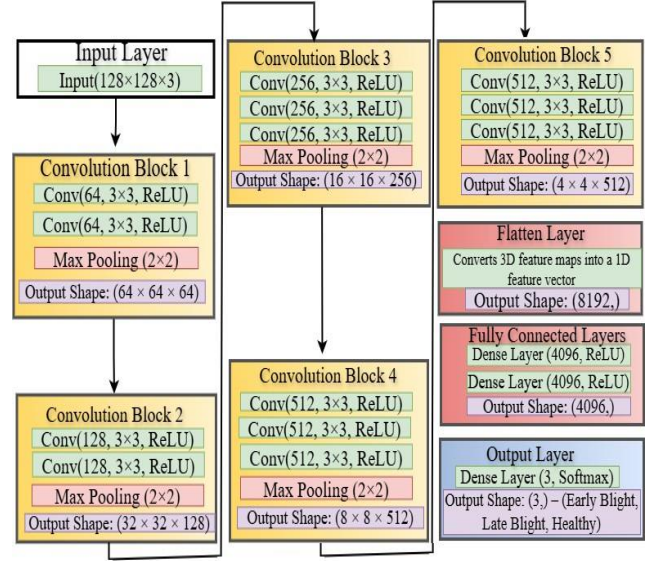


Fig. 2. VGG16 Architecture

At this point, the extracted features are changed into a one-dimensional form and sent to a fully connected (dense) layer. This layer helps the model combine the learned features and understand the overall pattern present in the image. After that, the data is passed to the final output layer, where the image is placed into the correct category using an activation function such as softmax, which works well for multi-class classification.

This design keeps the model simple while still giving good results. It does not require very high processing power, yet it maintains reliable accuracy. Because of this balance, the model can be used effectively for detecting plant diseases under real farming conditions.

B. VGG16 model

The VGG16 model is a deep learning architecture usually used for image classification tasks. In this study, RGB images of size $128 \times 128 \times 3$ are given as input and passed through multiple convolutional layers to extract useful features from the images. These layers use 3×3 filters along with ReLU activation, which helps the model learn important visual patterns. With the increase in depth of the network, the number of filters also increases from 64 to 512. This gradual increase permit the earlier layers to catch simple features, while the deeper layers focus on more difficult details. After passing through the last convolutional block, the feature maps are reshaped into one-dimensional vectors. These vectors are then sent to two fully connected layers, each consists 4096 neurons with ReLU activation, where deeper patterns are learned.

Using multiple convolutional and pooling layers helps improve disease detection accuracy. At the same time, pooling reduces extra information in the data, which helps reduces the chances of overfitting.

C. ResNet50 model

The ResNet-50 architecture used in this project is a deep convolutional neural network created to learn image features correctly using residual connections. As shown in Fig. 3, the model starts with an input layer that accepts images of size $128 \times 128 \times 3$.

After the input phase, the image passes through a first convolution layer followed by a pooling layer. This first convolution layer uses a 7×7 filter with 64 filters and a stride of 2. Batch normalization and ReLU activation are then applied to improve the learning process and maintain stable training. Following this, a 3×3 max pooling layer with a stride of 2 is used to reduce the size of the feature maps while keeping the important information.

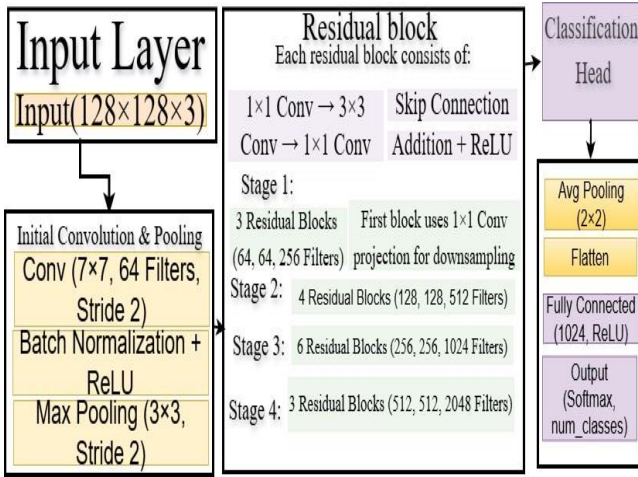


Fig. 3. ResNet50 Architecture

The network is created using residual blocks, which contains skip connections that help to decrease the vanishing gradient problem and allow the model to be trained more effectively at greater depth. The complete architecture is divided into four phase, with each phase becoming more difficult to improve feature extraction.

Stage 1 contains three residual blocks with 64, 64, and 256 filters. In this stage, the first block uses a 1×1 convolution for down sampling. Stage 2 includes four residual blocks with 128, 128, and 512 filters. Phase 3 further increases the depth and consists of six residual blocks with 256, 256, and 1024 filters. Finally, phase 4 contains three residual blocks with 512, 512, and 2048 filters, allowing the system to capture more information and high-level features.

After the feature extraction phase, a global average pooling layer (2×2) is applied to decrease the feature maps into a simpler form. The result is then goes to a completely connected layer with 1024 neurons using ReLU activation. In the last phase, a soft max layer is used to classify the images into the required types.

This system enhance the learning of main features and increases classification accuracy. Because of the deep residual connections, the model performs good in complex image recognition tasks.

D. InceptionV3

The The InceptionV3 model used in this study, as shown in Fig. 4, is a deep learning network designed to extract important image features efficiently and perform accurate classification. The model begins with an input layer that accepts RGB images of size $299 \times 299 \times 3$, which matches the required input size for this architecture.

The first part of the network is known as the Stem, which includes several convolutional layers. These layers mainly use 3×3 filters, along with stride-2 convolutions, to process the input images. A 3×3 max pooling layer with a stride of 2 is also included in this stage. This pooling step helps decrease the size of the images while still keeping the most important information.

By gradually decrease the image size and extracting useful features, this initial section helps the model analyze the images more effectively and improves the overall classification performance.

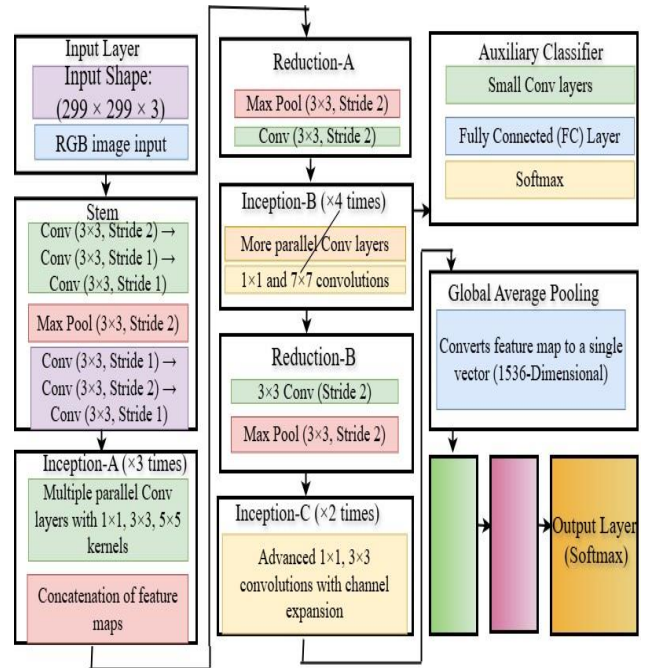


Fig. 4. InceptionV3 Architecture

The Inception V3 model utilizes specialized modules for feature extraction, specifically designed to capture patterns at various scales. The Inception-A module, repeated three times, performs multiple parallel convolutional layers using 1×1 , 3×3 , and 5×5 filters, merging the outputs to maintain diverse feature representations. To decrease the size of the feature maps while safeguarding key details, the Reduction-A module applies 3×3 max pooling and stride-2 convolutions. The Inception-B module runs four times and uses 1×1 and 7×7 convolutions to help the model see a larger area. This method helps to identify wider spatial relationships. The Reduction-B module uses 3×3 convolutions and max pooling with a stride of 2 to improve how we extract features. In the final phase, we apply Inception-C twice to enhance feature learning with advanced 1×1 and 3×3 convolutions, which increase the number of channels. After feature extraction, a global average pooling layer transforms the feature map into a single 1536-dimensional vector. Additionally, an auxiliary classifier, made of small convolutional layers, fully connected layers, and a softmax activation function, facilitates gradient flow during training. In conclusion, the model's output layer applies a softmax function to classify images into designated categories.

III. RESULTS

The performance of each deep learning model was checked using different classification metrics. These metrics helped in calculating how good each model identified potato leaf diseases. By evaluating the outcome, it became easier to understand the strengths and limitations of the four models used in this project.

A. Performance Comparison

The deep learning models performed very well in identifying plant leaf diseases, achieving an complete classification accuracy of 99%. The models were able to detect maximum diseases with high precision and recall values. However, the recall value for healthy leaves was slightly lower at 0.94, which means that a less healthy samples were incorrect classified as diseased.

Even with this minor limitation, the complete performance remained strong, with a weighted average score of 0.99 across all evaluation metrics. The outcome showed in Table 1 indicate that the model can truly detect potato diseases. This makes the model helpful for farmers, as it supports early disease detection and better crop management decisions

B. Model Accuracy Comparison

The accuracy comparison graph shows how the four deep learning models—Custom CNN, ResNet50, VGG16, and InceptionV3—performed during the training process. By observing the changes in training and validation accuracy, it becomes easier to understand how well each model learns and adjusts to the data. This comparison helps in identifying the strengths and weaknesses of each model and also shows their reliability in classifying leaf diseases.

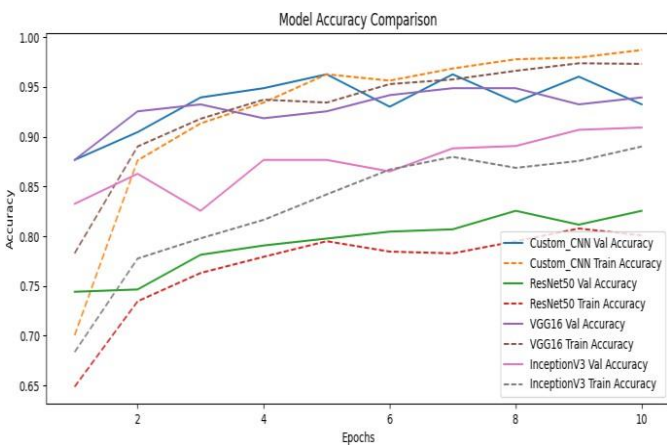


Fig. 5 Accuracy Comparison of Proposed Models

accuracy, The Custom CNN model shows strong learning ability, achieving high accuracy in both training and validation stages. The close values between these two accuracies suggest that the model has very little overfitting. The VGG16 model also performs steadily, with training and validation accuracy values that remain very close to each other. This indicates that the model is able to generalize well to new data.

The InceptionV3 model starts with lower accuracy in the starting, but its performance improves gradually over time. This steady improvement suggests that the model learns consistently before reaching stable accuracy levels. On the other hand, the ResNet50 model shows a noticeable gap between training and validation accuracy. This gap recommends possible overfitting or difficulty in adapting to the dataset, making it slightly less reliable compared to the other models in this project. data.

C. Confusion Matrix

The confusion matrix shows in Fig. 6 gives a clear view of how accurately the model classifies different potato leaf conditions. It contains both correct and incorrect predictions, which helps in understanding the complete performance of the model. In the matrix, the diagonal digits shows correctly classified samples, while the non-diagonal digits shows misclassified cases.

Even with these minor errors, the model manage strong classification performance. This represents that the system can correctly differentiate between different disease types and can be used as a helpful tool for predicting leaf diseases

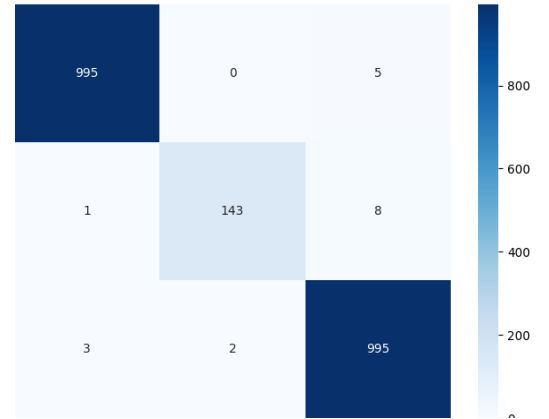


Fig. 6 Confusion Matrix

D. ROC Curve

The ROC curve represents in Fig. 7 helps defining how well the model differentiate between early blight, healthy, and late blight leaf classes. It represents the relationship between the True Positive Rate (TPR) and the False Positive Rate (FPR), which helps evaluate the classification performance of the model.

In this case, all three classes achieved an AUC value of 1.00, which indicates that the model classified the samples correctly with very high accuracy. The curve reaching close to the top-left corner represents that the model can correctly detect diseases while making very few false predictions.

Although these outcome show excellent work, such perfect scores may also suggest the possibility of overfitting. To ensure that the model works reliably, it is important to test it further using new and unseen data. This additional testing would help ensure that the model performs well in real-world conditions.

Deep Deep learning has significantly improved the process of plant disease detection and made it easier to classify crop infections with high accuracy. Among the models tested in this project, Custom CNN and VGG16 showed the best performance in identifying even very small disease patterns..Overall, these findings recommends that deep learning models can provide farmers with more reliable tools for automatic disease detection. Such systems can support early identification of plant diseases and help farmers take timely step to protect their crops. automatically.

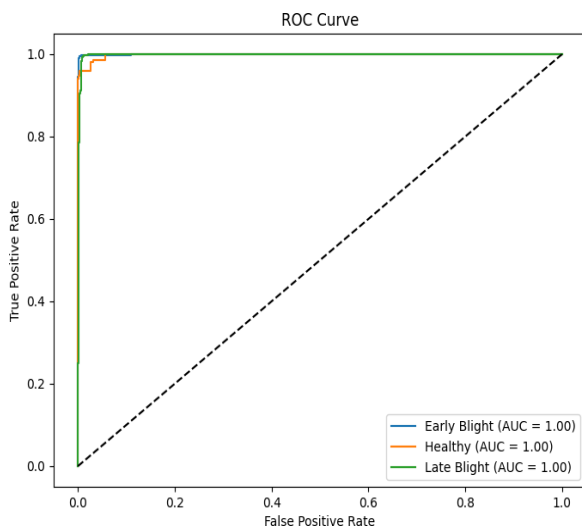


Fig. 7. ROC Curve

ResNet50 s The ResNet50 model showed slightly lower accuracy compared to the other models. This may be because it uses a deeper network, which can struggle when working with smaller agricultural datasets. Factors like high model complexity and gradient-related challenges can affect its working in such cases.

The InceptionV3 model performed relatively well, but it faced some difficulty in identifying very small changes in leaf symptoms. Because of this, its classification confidence was slightly lower than that of the top-performing models. These findings highlight the importance of selecting the most suitable model for plant disease detection tasks. They also show that more complex architectures usually require larger datasets and careful fine-tuning to achieve the best performance..

In addition to disease detection, the system provides practical support by including a pesticide recommendation feature. After detecting a disease, the system recommends suitable pesticides by considering factors like crop condition and previous treatment outcome. By combining disease detection with treatment suggestion, farmers can decrease unnecessary pesticide use, improve crop yield, and enhance more sustainable farming practices.

IV. CONCLUSION AND FUTURE SCOPE

This method helps make farming highly efficient and supports healthier crop growth while also decreasing harmful effects on the nature. The study also shows that explainable AI is very important in agricultural systems, not only for detecting diseases but also for helping farmers make better decisions in their daily farming work.

Farmers do not just need predictions from a model. They also need simple and clear explanations so they can understand what the system is explaining them and confirm whether the disease has been checked correctly. When the model define its decisions in an simple way, farmers feel more confident using the technology in real situations.

Overall, providing clear explanations makes disease management easier and more reliable. It also encourages responsible farming practices and helps farmers protect their crops in a more practical and effective way. practices.

In This research shows how artificial intelligence and deep learning can be useful in improving agricultural decision-making through an integrated advisory system. The proposed system combines plant disease detection, crop recommendation, and market price prediction to provide a complete support tool for farmers. By bringing these features together, the system becomes more practical for real farming needs.

Deep learning models like Custom CNN and VGG16 worked good in identifying plant diseases from leaf images. This makes disease detection faster and more reliable, which is important because early diagnosis can help prevent the spread of infection in crops. In addition to disease detection, the crop suggestions and price prediction models increase the usefulness of the system by helping farmers plan their crops and control resources more correctly.

The inclusion of pesticide and fertilizer suggestion also adds value by suggesting suitable treatments depends on the detected disease. This makes crop management easier and more focused. Overall, performing these tasks helps decrease crop damage, enhance productivity, and provides more efficient and sustainable farming methods in real agricultural environments. sector.

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